GeaNaVid

Gea, Na or Vid: verifying claims against Swiss federal voting booklets

Niklas Wichter

Team Apertool · Hack Apertus 2026 · Track 2A — OST: Multilingual NLI over Swiss Voting Booklets

- **Track:** Track 2A — OST: Multilingual NLI over Swiss Official Voting Booklets
- **Event:** Online
- **Team:** Apertool — Niklas Wichter
- **Demo:** [TBD: video link] · `make run` in `track_2a/`

Draft of 1 October 2026. All numbers are from runs on that day; open measurements are marked [TBD].

1. Summary

Swiss voters meet many claims about federal proposals; the official voting booklet is the authoritative source, but it runs to 40–90 pages in three languages. GeaNaVid checks a claim in German, French or Italian against a booklet and returns entailment, neutral or contradiction together with the passages that decide the label, the token usage and the time. We treat the task as document-level natural language inference (Koreeda & Manning, 2021; Yin et al., 2021), close to claim verification with evidence (Thorne et al., 2018), but judged only against the booklet, never against the world.

Our CLI `booklint` parses the PDF without a language model, retrieves a few passages with BM25 and asks Apertus v1.5 for a strict JSON verdict with evidence first and label second. Cross-lingual pairs are handled by searching the booklet's parallel version in the claim's language and mapping the hits back by page layout, which costs no tokens. On the official demo and its cross-lingual variants (74 pairs), Apertus 8B reaches a macro-F1 of 0.92 with about 2,200 input tokens per pair, against 0.81 with the whole booklet (18,200 tokens); Apertus 70B reaches 0.96 with retrieval and 0.98 with the whole booklet. On held-out synthetic claims for four longer booklets (40,000 tokens on average), the whole-booklet baseline drops to 0.60 (8B) and 0.85 (70B), while the ladder keeps 0.88 and 0.94 with 1,700–1,900 tokens. Our contributions are (1) a cost-aware context ladder for document NLI with evidence, (2) a measured trade-off between macro-F1, tokens and time for 8B and 70B, including the full-document baseline the challenge asks for, and (3) a zero-token cross-lingual retrieval step that uses the official parallel versions. The output is never a voting recommendation: every verdict comes with a quote from the booklet.

2. Architecture

The pipeline has five steps:

1. **Ingest** (once per booklet, no LLM, cached): text blocks with page, proposal, section and committee note.
2. **Claim language** by stop-word counts; no LLM.
3. **Retrieval**: BM25 top-5 plus same-page neighbours. Cross-lingual pairs search the parallel booklet in the claim language and map the hits back by layout; claims about a recommendation also get the recommendation blocks of the same proposal.
4. **Apertus NLI**, one call: numbered passages, JSON schema with evidence before label.
5. **Ladder gate**: an entailment or contradiction without evidence, or p(label) < 0.75, triggers one retry with the pages around the hits (±1 page). Output: label 0/1/2, evidence with page, section and offsets, tokens, seconds.

Ingest. PyMuPDF text blocks are cleaned (soft hyphens, line-break hyphenation, superscript footnote digits that otherwise glue to numbers). Each block keeps its page, the proposal from the running header ("Erste Vorlage: …", or an ordinal title in older layouts), the nearest bold side heading ("In Kürze", "Empfehlung von Bundesrat und Parlament") and the committee note ("Der Text auf dieser Doppelseite stammt vom Initiativkomitee"). The 40-page demo booklet gives 210 blocks in 0.3 s and zero tokens.

Retrieval. BM25 (Robertson & Zaragoza, 2009) over heading plus text, with number normalisation (500′000 = 500 000) and prefix stemming, fused across queries by reciprocal rank (Cormack et al., 2009). BM25 needs no GPU and is a strong zero-shot baseline (Thakur et al., 2021). For a cross-lingual pair we fetch the booklet in the claim's language (bk.admin.ch uses the same document id for all languages), rank there and map each hit to the block at the same relative position on the same page of the target booklet.

NLI call. The passages are numbered and prefixed with their context; Apertus answers under a JSON schema with the deciding passage numbers before the label, as in evidence-first setups (Koreeda & Manning, 2021; Thorne et al., 2018). Evidence is therefore always a real block of the booklet with page and character offsets, never generated text.

Ladder. The idea follows Self-Route and LLM cascades (Chen et al., 2023; Li et al., 2024): most pairs stop after one call; only unsure or unsupported verdicts climb to a wider context.

3. Use of Apertus

- **Model:** swiss-ai/Apertus-v1.5-8B and swiss-ai/Apertus-v1.5-70B (Project Apertus et al., 2025)
- **How it is used:** inference — the NLI verdict with evidence selection; optionally the translation of a claim (`--translate`).
- **Where it runs:** the Hack Apertus OpenAI-compatible endpoint, selected through `LLM_NAME`, `LLM_BASE_URL`, `LLM_API_KEY`; temperature 0, JSON schema, `logprobs` with 10 alternatives.

Both models honour the JSON schema and return log-probabilities, so the label probability is read from the first token of the label value. No other model takes part in the default path: retrieval is BM25 and parsing is deterministic. Prompts are versioned (`nli-v1` … `nli-v5`, Section 5.3); the default is `nli-v1`. Every call is cached on disk by a hash of

model, messages and parameters, which makes reruns free and reproducible.

4. Data

Set	Source	Pairs	Languages	E / N / C	Use
D0 demo	`OSTSwiss/MNLIoverSwissVotingBooklets`, commit 43f8aa3 (MIT)	23	3 decks, fr-fr, it-it	17 / 4 / 7	sanity, headline
D1 demo cross-lingual	D0 claims against the other two language versions	46	6 cross pairs	26 / 8 / 12	cross-lingual
D2 syn_dev	synthetic, 5 vote dates	132	same language	44 / 40 / 48	prompt and setting choice
D2 syn_test	synthetic, 4 other vote dates	101	same language	36 / 33 / 32	held out
D2 syn_x_test	syn_test claims against other language versions	120	cross	39 / 39 / 42	held out

The official demo has only 23 distinct claims from one booklet and no cross-lingual pairs. D1 carries the labels over to the official parallel versions and maps the gold evidence by page layout. D2 was generated for nine other votes (2009–2026) by two other LLMs (Claude and GPT-5.5 via the Codex CLI), each batch cross-reviewed by the other model; rows the review flagged were dropped. Apertus never generated or reviewed data. Generated claims cover hard cases on purpose: topical but unsupported, numbers and dates, attribution, negation, modality, recommendations (Nie et al., 2020; Schuster et al., 2021). Every row carries generator, model, date and prompt version; all rows still await a human check. The booklets are official publications of the Federal Chancellery; they are downloaded at run time and not stored in the repository. No personal data is involved.

5. Evaluation

Metrics. Macro-F1 over E/N/C with a 95 % bootstrap interval (1,000 resamples), F1 per class, retrieval recall (any retrieved block fuzzily inside the gold paragraph), grounded rate (correct E/C label and a cited block inside the gold paragraph), prompt tokens per pair from the API `usage`, LLM calls and seconds per pair.

5.1 Context versus macro-F1 (D0 + D1, 74 pairs)

Setup	8B macro-F1	70B macro-F1	Prompt tokens / pair
claim only (no booklet)	0.09	0.09	331
gold paragraph (oracle)	0.94	0.98	991
retrieval, top-5 + neighbours	0.89	0.96	1,690
retrieval + recommendation routing	0.91	0.96	1,717
ladder + routing (default)	**0.92** [0.84–0.98]	0.94 [0.88–0.99]	2,052–2,159
whole booklet	0.81 [0.70–0.91]	**0.98** [0.92–1.00]	18,229

The claim-only run predicts almost always neutral, so the claims carry no obvious label artefacts (Gururangan et al., 2018). For 8B, the whole booklet is worse than 9 % of it: retrieval removes distractors that the small model cannot ignore, in line with position effects in long contexts (Liu et al., 2024). For 70B the gap is 0.01–0.04 at a tenth of the tokens. Median latency per pair with retrieval is 0.34 s (8B) and 1.07 s (70B). Whole-booklet calls were not much slower (0.45 s and 0.86 s), probably because the endpoint caches the shared booklet prefix; we report this rather than claim a speed-up.

5.2 Cross-lingual retrieval (no LLM)

Retrieval	D1 recall@5	syn_x_test recall@5
BM25 on the target booklet with the original claim	0.74	0.42
BM25 in the parallel booklet, mapped back by layout	**0.95**	**0.88**
same-language reference (D0 / syn_dev)	0.96	0.97

The gold evidence of D1 and syn_x_test was itself mapped by layout, which favours the mapping; a comparison with Apertus translation of the claim (Artetxe et al., 2023) is [TBD].

5.3 Harder synthetic cases and prompt variants (syn_dev, 132 pairs, 8B)

Prompt	Change	Retrieval	Ladder
nli-v1	label definitions, attribution check	**0.873**	**0.896**
nli-v2	"a missing detail is neutral", stricter attribution	0.840	0.862
nli-v4	v1 + booklet conventions (Federal Council and Parliament, "voting text")	0.858	0.872
nli-v5	v4 + one-sentence check before the label	0.845	0.878

With 70B and nli-v1 the same set gives 0.963 (oracle), 0.933 (retrieval) and 0.948 (ladder). The ladder helps on hard cases (+0.02 for both models). None of the prompt variants beats v1; v2 moved true contradictions (numbers, negations) to neutral. Calibrating a neutral bias on the label log-probabilities (Zhao et al., 2021) did not help either: more than 90 % of the answers have p(label) > 0.9, so the probability gate fires for only about 2 % of pairs and the ladder mostly escalates verdicts without evidence.

5.4 Held-out synthetic test

Settings were frozen on syn_dev before these runs; each held-out set was run once.

Set	Setup	Prompt tokens / pair	8B	70B
syn_test (101)	gold paragraph (oracle)	484	0.940 [0.89–0.98]	0.930 [0.87–0.97]
syn_test (101)	retrieval	1,446	0.881 [0.81–0.94]	0.931 [0.88–0.97]
syn_test (101)	ladder	1,715–1,856	0.882 [0.81–0.94]	**0.941** [0.89–0.98]
syn_test (101)	whole booklet	40,149 (p95 69,689)	0.604 [0.50–0.70]	0.847 [0.77–0.91]
syn_x_test (120)	gold paragraph (oracle)	488	0.851 [0.79–0.91]	0.859 [0.79–0.92]
syn_x_test (120)	retrieval	1,413	**0.857** [0.79–0.92]	**0.899** [0.84–0.95]
syn_x_test (120)	ladder	1,807–1,832	0.854 [0.78–0.91]	0.882 [0.82–0.93]

The held-out booklets (2010, 2016, 2020, 2025) are longer than the demo, with up to five proposals. Here the whole-booklet baseline breaks down: 8B falls to 0.60 and 70B to 0.85, while retrieval and the ladder keep 0.88 and 0.93–0.94 with 4 % of the tokens. Retrieval stays within 0.06 (8B) and 0.01 (70B) of the gold paragraph. Whole-booklet latency for 70B rises to 2.8 s median and 16 s at p95, against 0.9 s for the ladder. Cross-lingual pairs cost 0.02–0.06 macro-F1 against the same claims in their own language. That the oracle is not better than retrieval there points to the layout-mapped gold paragraphs of this set being noisy. The ladder helps 8B on syn_dev and 70B on syn_test, but not on the cross-lingual set; with 70B, escalation sometimes turns a correct verdict into a wrong one.

Error analysis (8B, syn_dev, retrieval). Of 17 errors, 6 are topical but unsupported claims judged as contradiction (a figure the booklet never gives), 5 are attribution errors (the committee says X, the Federal Council the opposite, and the model follows the committee text), 3 are number contradictions judged as entailment. On the demo, 8B misread a paraphrased legal threshold ("nine and a half million") as a contradiction in three language versions of the same claim.

Evidence. Grounded rate against the single gold paragraph is 0.68–0.73 with retrieval and 0.39–0.45 with the whole booklet. Many misses cite another valid passage (for example the legal text instead of the summary), so this is a lower bound; a judged evaluation of evidence relevance is [TBD].

6. Limitations

The development data are small: 23 distinct official claims from one booklet, and the cross-lingual and synthetic sets are our own, with labels not yet checked by a human; at $n \approx 100$ a macro-F1 difference below about 0.08 is within noise. Synthetic claims may follow their generators' style. Topical but unsupported claims and attribution remain the main error sources for 8B. The parallel-booklet step needs the bk.admin.ch URL; for a local PDF it falls back to direct BM25 or translation. Scanned booklets without a text layer (before about 2000) are not handled yet. Labels and evidence are an automated reading of the booklet, not a voting recommendation, and the model's position must never be presented as the booklet's.

7. Reproducibility

`make run` builds the Docker image and runs the benchmark on D0 + D1 against the endpoint in `LLM_*`; no GPU or local weights are needed. Settings: prompt `nli-v1`, $k = 5$, one neighbour, $\tau = 0.75$, temperature 0, JSON schema. Development ran on a laptop against the Hack Apertus endpoint; commit [TBD]. Runs write one JSONL per setting with label, evidence, probabilities, tokens and seconds, plus `summary.md`.

8. Next steps

Measure Apertus translation against the parallel-booklet step; add OCR text for scanned booklets (Stüssi & Vamvas, 2026); human-check the synthetic sets and judge evidence relevance; try 70B only for escalated pairs; extend to Romansh.

License

Creative Commons Attribution 4.0 (CC-BY-4.0). All HackApertus projects are open-sourced.

References

Artetxe, M., Goswami, V., Bhosale, S., Fan, A., & Zettlemoyer, L. (2023). Revisiting Machine Translation for Cross-lingual Classification. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 6489–6499. https://doi.org/10.18653/v1/2023.emnlp-main.399

Chen, L., Zaharia, M., & Zou, J. (2023,). *FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance*. https://arxiv.org/abs/2305.05176

Cormack, G. V., Clarke, C. L. A., & Buettcher, S. (2009). Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods. *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 758–759. https://doi.org/10.1145/1571941.1572114

Gururangan, S., Swayamdipta, S., Levy, O., Schwartz, R., Bowman, S. R., & Smith, N. A. (2018). Annotation Artifacts in Natural Language Inference Data. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, 107–112. https://doi.org/10.18653/v1/N18-2017

Koreeda, Y., & Manning, C. (2021). ContractNLI: A Dataset for Document-level Natural Language Inference for Contracts. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 1907–1919. https://doi.org/10.18653/v1/2021.findings-emnlp.164

Li, Z., Li, C., Zhang, M., Mei, Q., & Bendersky, M. (2024). Retrieval Augmented Generation or Long-Context LLMs? A Comprehensive Study and Hybrid Approach. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 881–893. https://doi.org/10.18653/v1/2024.emnlp-industry.66

Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the Middle: How Language Models Use Long Contexts. *Transactions of*

the Association for Computational Linguistics, *12*, 157–173. https://doi.org/10.1162/tacl_a_00638

Nie, Y., Williams, A., Dinan, E., Bansal, M., Weston, J., & Kiela, D. (2020). Adversarial NLI: A New Benchmark for Natural Language Understanding. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4885–4901. https://doi.org/10.18653/v1/2020.acl-main.441

Project Apertus, Hernández-Cano, A., Hägele, A., & others. (2025,). *Apertus: Democratizing Open and Compliant LLMs for Global Language Environments*. https://arxiv.org/abs/2509.14233

Robertson, S., & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, *3*, 333–389. https://doi.org/10.1561/1500000019

Schuster, T., Fisch, A., & Barzilay, R. (2021). Get Your Vitamin C! Robust Fact Verification with Contrastive Evidence. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 624–643. https://doi.org/10.18653/v1/2021.naacl-main.52

Stüssi, E., & Vamvas, J. (2026). How Good is AI on Swiss Voting Booklets? A Multilingual OCR and Alignment Benchmark. *Proceedings of the 11th Edition of the Swiss Text Analytics Conference*, 232–242. https://aclanthology.org/2026.swisstext-1.22/

Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., & Gurevych, I. (2021,). BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models. *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*.

Thorne, J., Vlachos, A., Christodoulopoulos, C., & Mittal, A. (2018). FEVER: a Large-scale Dataset for Fact Extraction and VERification. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, 809–819. https://doi.org/10.18653/v1/N18-1074

Yin, W., Radev, D., & Xiong, C. (2021). DocNLI: A Large-scale Dataset for Document-level Natural Language Inference. *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 4913–4922. https://doi.org/10.18653/v1/2021.findings-acl.435

Zhao, T. Z., Wallace, E., Feng, S., Klein, D., & Singh, S. (2021). Calibrate Before Use: Improving Few-Shot Performance of Language Models. *Proceedings of the 38th International Conference on Machine Learning (ICML), Proceedings of Machine Learning Research*, *139*, 12697–12706.